

Neuro-Evolved Heuristics for Variable Gapped Common Subsequence Identification

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Introduction

- Objects we deal with: sequences (strings) over finite alphabet Σ
 - DNA/RNA sequences over $\{A, T, G, C/U\}$
 - Proteins over 20 (canonical) amino acids: $\{A, C, D, E, P, Q \dots\}$
- One of central tasks in computational biology:
 - sequence comparison, finding **common motifs** between sequences
 - how structural similarity affects sequence functionality
- Motifs can be **subsequences**: **Longest common subsequence problem** variants (studied 50 years already)

Longest common subsequence problem (LCSP)

Definition (LCSP)

Input: Given an arbitrary set of sequences $S = \{s_1, \dots, s_m\}$

Task: Find a subsequence s **common** for all sequences from S with **maximum** possible length ($|s|$).

LCS: Literature & Problem Variants

- For $m = 2$, polynomially solvable, e.g. Dynamic programming
- When m arbitrary large, it is **NP-hard**: approximation, meta-heuristic, exact approaches

Problem-related practical variants: Repetition-free, Filled LCS problem, **Gapped LCS problem**

The gapped LCS problem (Peng and Yang, 2012)

Definition (Gapped sequence)

Given is a sequence s and a function $G_s: \{1, \dots, |s|\} \mapsto \mathbb{N}$. A pair (s, G_s) is called a gapped sequence.

(G_s : To each position of seq. s , one number assigned)

Definition (Gapped subsequence)

Sequence $\tilde{s}$ is a **gapped subsequence** of (s, G_s) iff

- $\tilde{s}$ is a **subsequence** of s
- the gap constraint G_s fulfilled resp. the *positions of embedding* $\tilde{s}$ in s :
 - consecutive embedding positions respect gap distances (no more than the **assigned number of letters** inbetween respected)

Problem definition

Example

$s = \text{AATTGC}$, $G_s(\cdot) = 1$, $\tilde{s} = \text{ATG}$

- the embedding **AATTGC** (**valid** gapped subsequence)
- the embedding **AATTGC** (**invalid** gapped subsequence)

Definition (The multiple (variable) gapped LCS problem – MVGLCSP)

Input: Given is a set of m gapped sequences $\{(s_i, G_{s_i})\}_{i=1}^m$.

Task: Find the longest sequence $\tilde{s}$ so that

- $\tilde{s}$ is **common subsequence** of each s_i
- $\tilde{s}$ **fulfills all gap constraints** G_{s_i} ($i = 1, \dots, m$)

Literature & Motivation for VGLCSP

- Peng and Yang (2012, 2014): $m = 2$ version by **three dynamic programming** (DP) approaches
- Manea et al. (2024): complexity analysis investigated
 - **NP-hard** under arbitrary large m
- Djukanovic et al. (2026): the first approach for m arbitrary: **Iterative Beam search approach**

Motivation:

- **Genetics and molecular biology**: DNA/protein comparison analysis—distances between residues must be respected
- **Time-series analysis**: in settings where events are required to occur within specified temporal delays (Lainscsek et al. (2015))

Methodology by Djukanovic et al. (2026)

- Based on the **state space graph formulation** of the problem
- **Nodes** corresp. to partial solutions and resp. subproblems to be solved
 - transitions: valid extension of partial solutions (by one letter)
- **Gap constraints**: incorporated to cut-off invalid extensions (edges) among the LCS extensions immediately
- **Many root/source nodes** in the state space—the position of leading letter in each common gapped subsequence free to select
 - root state space subgraph (formulation) w.r.t. starting source (match)

Methodology: main components roughly

- Generation of longest solution for a root (sub-)space is generally hard
⇒ need an approx. approach for evaluation (**Beam search – limited BFS**)
- The root-state-space formulation may involve disconnected components generation ⇒ needs **iterative beam search** to explore all space

⇒ **Iterative Multi Source Beam Search (IMSBS) approach.**

Heuristic guidances in IMSBS

The best literature heuristic to guide BS and prioritize root node:

- ① UB_2 : *Character Frequency Alignment* score

Experimental evaluation:

- IMSBS able to efficiently run on large synthetic instances
- **Weaknesses:** the performance of UB_2 seems to significantly drop for large n ($n = 500$) and small $|\Sigma|$ ($|\Sigma| = 2$)

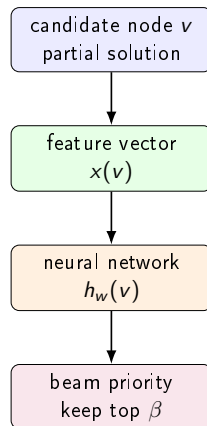
⇒ **Construction of a new heuristic guidance**

Designing advanced heuristic guidances: main idea

- IMSBS needs a better score to decide which candid nodes stay in the beam
- We learn a **new node evaluator**: a function assigning a promise score to a partial solution

Role of the learned heuristic

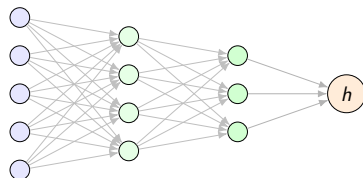
It does **not** construct the solution directly; it ranks candidate states during search.



Neural network basics: MLP as a scoring function

- We use a **feed-forward multilayer perceptron** (MLP).
- Input: **compact numerical description** of the state v .
- Hidden layers transform features into nonlinear combinations.
- Output: a **single heuristic score** $h_w(v)$.

features hidden + activation funcs score



State features for VGLCS node evaluation

For a state $\mathbf{v} = (p^v, l^v)$ (p^v : vector of suffix positions of subproblem to be solved, l^v : length of partial solution), build a size-independent features:

Search progress

- normalize positions:
 $F_i^v = p_i^v / |s_i|$
- max, min, mean, std of F^v
- current length l^v

Gap flexibility

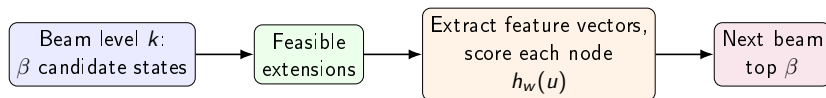
- $G_i^v = G_{s_i}[p_i^v, \dots]$
- max, min, mean, std
- describes how restrictive future extensions are

Global context

- alphabet size $|\Sigma|$
- number of sequences m
- standardized before training

Feature sets tested: $F_1 = \text{progress} + \text{gaps} + \text{length}$ (9); $F_2 = F_1 + |\Sigma|$ (10);
 $F_3 = F_2 + m$ (11).

How the learned evaluator is used inside IMSBS



Important note

Learning is **offline**; during the final run the NN is only a fast scoring function.

Why not standard supervised backpropagation?

Obstacle

- Exact optimal trajectories unavailable for the general case
- Search quality non-differentiable: one ranking decision affect many later states

Solution: neuro-evolution

- Treat all NN weights as **one vector** w .
- Evaluate w by running (inexpensive) IMSBS on training (32) instances.
- **Fitness** = average subsequence length found

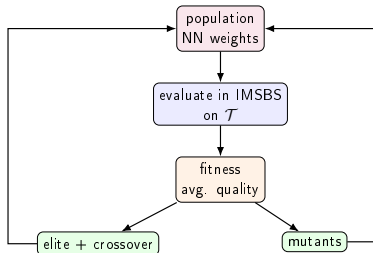
$$\max_w \frac{1}{|\mathcal{T}|} \sum_{I \in \mathcal{T}} f(\text{IMSBS}(I, h_w))$$

Learning weights with Biased Random Key GA

- Individual = complete NN weight vector w .
- Population of weights, evolution
- ...

Parameters used

$n_{pop} = 20$, $n_{elite} = 1$, $n_{mutants} = 7$,
 $p_{bias} = 0.5$.



Rank-based ensemble: learned + analytical guidance

- Preliminary tests: the learned heuristic strong, but not uniformly best
- Hand-crafted upper bounds useful, especially in some small- m cases
- Raw scores not directly comparable, **rankings** rather than scores combined

Combined rank

$$r(v) = \alpha r_{h_{w^*}}(v) + (1 - \alpha) r_{UB_2}(v).$$

Lower rank $r(v)$ means higher priority in the beam.

⇒ UB_2 replaced by $r(\cdot)$: **Ensemble-IMSBS approach**.

What was tuned? Experimental design

IMSBS parameters

- tuned by irace (parameters from the literature: $\beta = 5000$)

Training protocol

- IMSBS utilized $\beta = 500$, max. 100 beam iterations or 60 sec

NN design choices

- 2 or 3 hidden layers
- 5 or 10 neurons per hidden layer
- activation: tanh, ReLU, sigmoid
- features: F_1 , F_2 , F_3

Selected architectures (split by diff. m)

m	layers	act.	features
2	10-5-5	ReLU	F_2
3	10-10-5	ReLU	F_2
5	10-5-5	sigmoid	F_1
10	10-5-5	ReLU	F_2

Numerical results: Random and Real benchmarks

Random benchmark

comparison [Ensemble vs. IMSBS]	wins	ties	losses
all groups (32)	20	8	4
$m = 2, 3$	10	5	1
$m = 5, 10$	10	3	3

Wilcoxon: $p = 0.009$ on all groups.

Real benchmark

comparison	wins	ties	losses
LIMSBS-Ensemble vs. IMSBS (20)	12	7	1

Comparable runtime; larger gains on harder instances such as Virus.

Takeaway

The learned ensemble improves solution quality consistently, while keeping the original IMSBS machinery almost unchanged.

How close are we to the exact DP-1 solution ($m = 2$)?

$$|\Sigma| = 2$$

n	Avg. Gap
50	1.0%
100	6.5%
200	10.0%
500	24.4%

Performance starts to degrade
for large instances.

$$|\Sigma| = 4$$

n	Avg. Gap
50	0.7%
100	0.3%
200	1.1%
500	2.0%

Near-optimal performance
across all instance sizes.

Ensemble-IMSBS stays within 2% of DP-1
for all tested $|\Sigma| = 4$ instances.

Conclusion

- Learning an improved guidance promising (best-performing heuristic)
- Low alphabet diversity, requires even stronger exploration/**richer feature portfolio**.